

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab3

Student Name: Samira Donkoh

Student ID: 20962028

GitHub Repository Link: https://github.com/miradonkoh/lab_3.git

Date Submitted: 28/07/2026

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	Yes
Estimated percentage of the work influenced by AI	20%
Did you attach evidence of AI use where applicable?	yes

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
Chat gpt	To correct my errors, and explain certain stuff to me		Errors I had.	I changed some of the implementations given, and for some I directly copied the correct code.

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft		
Prompt history		
Screenshot of AI interaction		

Exported AI conversation		
AI-generated code snippet		
Revised version showing student input		

4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is my own work.	Yes
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes
My AI usage does not exceed 25% of the entire work.	Yes
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

Student Signature:samiradonkoh

Date: 28/07/2028

this is my code, and im getting this error, # TODO: Initialise parameters

```
# W1: shape (1, 8), b1: shape (8,)
# W2: shape (8, 1), b2: shape (1,)
# Small random values (e.g. np.random.randn(...) * 0.5)
W1=np.random.randn(1,8)*0.5
b1=np.random.randn(8)*0.5
W2=np.random.randn(8,1)*0.5
b2=np.random.randn(1)*0.5
lr=0.01
X=x.reshape(-1,1)
Y=y.reshape(-1,1)
```

Show more ▾

The error is happening at:

Python



Run

```
dh = d_yhat @ W2.T
```

The shapes of your matrices do not match for matrix multiplication.

Let's check your shapes.

Forward pass



You have:

+ Ask anything



```
i did thi from google.colab import drive
drive.mount('/content/drive')
import pandas as pd

OBESITY_URL = "Lab2/Obesity_level_prediction_dataset (2).csv"
obesity = pd.read_csv(OBESITY_URL) and got this error Mounted at
/content/drive
-----
FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_1972/3202053899.py in <cell line: 0>()
      5

Show more
```



The error means that Colab cannot find the file at that location. Mounting Google Drive does not automatically make "Lab2/..." a valid path.

Let's find the correct path.

Step 1: Check what's in your Google Drive

Run this:

```
Python
import os

os.listdir("/content/drive/MyDrive")
```

If you see a folder called `Lab2`, then your file is probably here:

+ Ask anything



there was no output when i run it

If `!find /content/drive/MyDrive -name "*obesity*"` produced **no output**, then the dataset is **not in your Google Drive** (or it's named something completely different).

Let's check where the file actually is.

Option 1: Check the Files panel in Colab

On the left side of Colab, click the **folder icon** .

- Do you see `Obesity_level_prediction_dataset (2).csv`?
- Do you see a folder named `Lab2`?

If yes, tell me what you see.

Option 2: List the current directory

Run:

```
Python
import os

os.listdir()
```

and also:

```
Python
!pwd
```

+ Ask anything



```
now im getting this error File "/tmp/ipykernel_1972/968920395.py",
line 6
import torch.utils.data import TensorDataset, DataLoader
                        ^
SyntaxError: invalid syntax
```

The error is because the import statement has the wrong syntax.

You wrote:

```
<< Python
import torch.utils.data import TensorDataset, DataLoader
```

Python does not allow `import` twice in that format.

Use this instead:

```
<< Python
import torch
from torch.utils.data import TensorDataset, DataLoader
```

So your full import section should be:

```
<< Python
import pandas as pd
import numpy as np
```

```
now this error AssertionError                                Traceback (most recent
call last)
/tmp/ipykernel_1972/3316763579.py in <cell line: 0>()
      55 traindataset=TensorDataset(Xtraintensor,ytraintensor)
      56 valdataset=TensorDataset(Xvaltensor,yvaltensor)
--> 57 testdataset=TensorDataset(Xtesttensor,yvaltensor)
      58
      59
train_loader=DataLoader(traindataset,batch_size=32,shuffle=True)

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataset.py in
Show more ▾
```

This error is because your **test features** and **test labels** have **different numbers of rows**.

The problem is this line:

```
<< Python
testdataset=TensorDataset(Xtesttensor,yvaltensor)
```

You are pairing:

- `xtesttensor` → test data (20% of dataset)
- `yvaltensor` → validation labels (also 20%, but a **different split**)

Even though they may have the same size in this case, Torch is detecting they do not match.

You need to use the **test labels**:

+ Ask anything



```
im getting errors here # TODO: criterion = nn.CrossEntropyLoss()
# optimizer = torch.optim.Adam(model.parameters(), lr=1e-3) # or
SGD — justify
criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=1e-3)

# TODO: Train for at least 50 epochs. Each epoch:
# 1. model.train(); for xb, yb in train_loader:
# optimizer.zero_grad()
# logits = model(xb.to(DEVICE))
# loss = criterion(logits, yb.to(DEVICE))

Show more ▾
```

The error is caused by a **typo**:

You created:

Python

```
logits = model(xb)
```



Run

but then you wrote:



Python

+ Ask anything



Run



```
/tmp/ipykernel_1972/2893314729.py in <cell line: 0>()
    52     predicted=torch.argmax(logits,dim=1)
    53     total+=yb.size(0)
--> 54     valcorrect+=(predicted==yb).sum().item()
    55     valloss=valloss/len(val_loader)
    56     valaccuracy=valcorrect/total
```

NameError: name 'valcorrect' is not defined now im gettin this

This error is because you are using `valcorrect` here:

Python



Run

```
valcorrect += (predicted == yb).sum().item()
```

but you never created the variable before the validation loop.

You need to initialize it at the start of your validation section.

Find this part:

Python



Run

```
model.eval()
valloss=0
correct=0
total=0
```



Change it to:

+ Ask anything

